

MINIMUM IDENTIFYING FEATURES FOR HISTOLOGY SPOTTING

1. HYALINE CARTILAGE

- a. Presence of perichondrium – fibrous and cellular layer
- b. Chondrocytes in cell nests within lacuna
- c. Homogenous matrix with metachromasia (Territorial and interterritorial matrix)

2. ELASTIC CARTILAGE

- a. Presence of perichondrium – fibrous and cellular layer
- b. Large chondrocytes within lacuna arranged singularly
- c. Elastic fibres within the matrix

3. FIBROCARILAGE

- a. Absence of perichondrium
- b. Chondrocytes arranged in linear manner between parallel bundles of collagen fibres

4. BONE Transverse section

- a. Haversian system with central Haversian Canal
- b. Osteocytes in lacunae arranged in between concentric, interstitial and circumferential lamellae and connected by canaliculi between lamellae

5. BONE Longitudinal section

- a. Longitudinal sections of Haversian and Volkmann's canal seen
- b. Osteocytes in lacunae arranged parallel to Haversian canal.

6. SKELETAL MUSCLE

- a. Unbranched long fibres with dark & light cross striations.
- b. Syncytium, sarcoplasm with peripheral flattened nucleus

7. CARDIAC MUSCLE

- a. Branched fibres with faint cross striations.
- b. Central ovoid nucleus with intercalated discs separating adjacent myocytes.

8. SMOOTH MUSCLE

- a. Spindle shaped cells with broad central part and tapering ends.
- b. Single ovoid or elongated central nucleus in each muscle cell.

9. ELASTIC ARTERY

- a. Tunica intima, Tunica media and Tunica adventitia .
- b. Prominent Tunica media containing abundant elastic fibres arranged in a concentric manner

10. MUSCULAR ARTERY

- a. Tunica intima, Tunica media and Tunica adventitia.
- b. Prominent Tunica media with smooth muscle cells arranged in a concentric manner
- c. Prominent internal elastic lamina

11. LARGE VEIN

- a. Tunica intima, Tunica media and Tunica adventitia
- b. Prominent Tunica adventitia with longitudinally arranged smooth muscle fibres

12. DORSAL ROOT GANGLIA

- a. Large round to ovoid cell bodies of pseudounipolar neurons arranged in groups separated by bundles of myelinated nerve fibres.
- b. Numerous satellite cells surrounding each nerve cell.

13. AUTONOMIC GANGLIA

- a. Small, irregular Multipolar cell bodies with eccentrically placed nucleus scattered in between thin unmyelinated nerve fibres
- b. Few satellite cells
- c.

14. OPTIC NERVE

- a. Covered by Duramater, Arachnoidmater and Piamater
- b. Central artery and vein of retina
- c. Piamater divides nerve substance into polygonal fasciculi

15. THIN SKIN

- a. Epidermis lined by stratified squamous keratinized epithelium
- b. Numerous hair follicles, sebaceous glands and arrectores pilorum
- c. Papillary and reticular layer of Dermis

16. THICK SKIN

- a. Epidermis lined by stratified squamous keratinized epithelium with prominent Stratum Corneum and Lucidum
- b. Absence of hair follicles
- c. Papillary and reticular layer of Dermis

17. LYMPH NODE

- a. Capsule with subcapsular sinus
- b. Dark staining outer cortex with lymphocytes arranged in the form of lymphatic nodules with pale staining germinal centre and inner medulla
- c. Pale staining inner medulla with lymphocytes arranged in form of Medullary sinus

18. THYMUS

- a. Incomplete lobulation with lymphocytes arranged as outer cortex and inner medulla
- b. Hassall's corpuscle in medulla

19. SPLEEN

- a. Lymphocytes arranged as red pulp and white pulp
- b. Red pulp with lymphocytes and Red blood corpuscles interspersed
- c. White pulp with lymphocytes arranged around eccentrically placed arteriole

20. TONSIL

- a. Tonsillar crypts lined by stratified squamous nonkeratinised epithelium
- b. Lymphocytes arranged in the form of nodules.

21. PLACENTA

- a. Sections of irregular chorionic villi, lined by cytotrophoblast and syncytiotrophoblast
- b. Maternal RBC in intervillous space
- c. Hofbauer cells seen

22. UMBILICAL CORD

- a. Core of mucoid connective tissue (Wharton's jelly) covered by amnion
- b. Section of two umbilical arteries and one umbilical vein.

23. MUCOUS SALIVARY GLAND

- a. Mucous acini of various shapes, large lumen, lined by columnar cells having pale foamy cytoplasm, flat basal nuclei.
- b. Interlobular, Intralobular and intercalated ducts lined by columnar, cuboidal and squamous epithelium

24. SEROUS SALIVARY GLAND

- a. Rounded serous acini, small lumen, pyramidal cells with centrally placed rounded nuclei bipolar staining cytoplasm (basal basophilia and apical eosinophilia)
- b. Interlobular, Intralobular and intercalated ducts lined by columnar, cuboidal and squamous epithelium.

25. MIXED SALIVARY GLAND

- a. Lobes made up of serous and mucous acini
- b. Crescent shaped groups of serous cells capping the mucous acini (Demilunes of Heidenhain or crescents of Gianuzzi)

- c. Inter lobular, Intralobular and intercalated ducts lined by columnar, cuboidal and squamous epithelium.

26. PANCREAS

- a. Exocrine part- serous acini with bipolar staining, centroacinar cells
- b. Endocrine part – pale staining Islets of Langerhans

27. LIVER

- a. Hepatic lobule with central vein and hepatocytes radiating from it. Hepatocytes separated by sinusoids
- b. Portal triad with cut sections of portal vein, hepatic artery, bile duct

28. GALL BLADDER

- a. Mucosa, fibromuscular coat, serosa.
- b. Simple tall columnar epithelium with microvilli.
- c. Smooth muscle fibres interspersed with elastic fibres are seen in the fibromuscular coat.

29. TONGUE

- a. Lining epithelium is stratified squamous non-keratinised.
- b. Presence of papillae-Filiform, fungi form and circumvallate papillae
- c. Central core of fibromusculoglandular tissue

30. ESOPHAGUS

- a. Mucosa, submucosa, muscularis externa & adventitia
- b. Mucosa lined by nonkeratinised stratified squamous epithelium thrown into folds.
- c. Prominent submucosa with tubuloacinar mucous glands.

31. STOMACH -FUNDUS

- a. Mucosa, submucosa, muscularis externa and serosa.
- b. Mucosa – lined by simple columnar epithelium , gastric pits extending into $\frac{1}{4}^{\text{th}}$ of the mucosa, Lamina propria with numerous gastric glands - Acidophilic parietal cells and basophilic chief cells seen.

32. STOMACH- PYLORUS

- a. Mucosa, submucosa, muscularis externa and serosa.
- b. Mucosa – lined by simple columnar epithelium with gastric pits extending into $\frac{3}{4}^{\text{rd}}$ of the mucosa. Pyloric mucous glands are seen in lamina propria.

33. DUODENUM

- a. Mucosa, Submucosa, Muscularis externa and serosa
- b. Mucosa –finger like villi lined by columnar cells with striated border and goblet cells.
- c. Mucous Brunner's glands in submucosa.

34. JEJUNUM

- a. Mucosa, Submucosa, Muscularis externa and serosa
- b. Mucosa –leaf like villi lined by columnar cells with striated border and goblet cells.
- c. Featureless submucosa.

35. ILEUM

- a. Mucosa, Submucosa, Muscularis externa and serosa
- b. Rudimentary villi lined by columnar cells with striated border and goblet cells.
- c. Peyer's patches- lymphoid aggregates.

36. LARGE INTESTINE

- a. Mucosa, Submucosa, Muscularis externa and serosa
- b. Absence of intestinal villi
- c. Crypts of Leiberkuhn with numerous goblet cells

37. APPENDIX

- a. Small lumen with Mucosa, Submucosa, Muscularis externa and serosa
- b. Crypts of Leiberkuhn lined by simple columnar epithelium with goblet cells

- c. Lamina propria - lymphoid aggregates.

38. KIDNEY

- a. Capsule, cortex, medulla
- b. Cortex shows sections of renal corpuscles, darkly stained PCT (small lumen, lined by a single layer of large cuboidal cells with eosinophilic granular cytoplasm and brush borders) and pale stained DCT (larger lumen lined by smaller cuboidal cells)
- c. Medulla shows collecting tubules and ducts and Loop of Henle (Thin - simple squamous epithelium, thick - simple cuboidal epithelium)

39. URETER

- a. Mucosa, muscular coat, adventitia
- b. Star shaped lumen lined by transitional epithelium
- c. Muscular layer showing outer circular and inner longitudinal layer of smooth muscle

40. URINARY BLADDER

- a. Mucosa, muscular layer, serosa /adventitia
- b. Folded mucous membrane lined by transitional epithelium
- c. Muscular wall showing 3 layers inner & outer longitudinal, middle circular layer of smooth muscle.

41. UTERUS – SECRETORY PHASE

- a. Endometrium, myometrium, perimetrium
- b. Endometrium lined by simple columnar epithelium and differentiated functionally into stratum compactum, stratum spongiosum, stratum basale.
- c. Cut sections of dilated, tortuous coiled uterine and coiled arteries (corkscrew appearance).
- d. Myometrium shows thick layer of smooth muscle fibre bundles interlacing in many directions and connective tissue

42. UTERUS – PROLIFERATIVE PHASE

- a. Endometrium, myometrium, perimetrium
- b. Endometrium lined by simple columnar epithelium with long straight tubular glands.
- c. Myometrium shows thick layer of smooth muscle fibre bundles interlacing in many directions and connective tissue.

43. OVARY

- a. Cuboidal germinal epithelium with inner Tunica albuginea
- b. Outer cortex with numerous ovarian follicles in various stages of development (primordial, primary, secondary, tertiary, mature)
- c. Medulla - stroma of irregular connective tissue with blood vessels.

44. FALLOPIAN TUBE

- a. Mucous, muscular, serous coat
- b. Mucosa thrown into complex folds - primary secondary & tertiary – lined by ciliated simple columnar epithelium

45. MAMMARY GLAND -ACTIVE

- a. Tubuloalveolar glands and connective tissue stroma
- b. Glandular tissue predominates, distended alveoli lined by cuboid cells filled with milk secretions, ducts containing secretion
- c. Stroma ill-defined

46. MAMMARY GLAND - INACTIVE

- a. Tubuloalveolar glands and connective tissue stroma
- b. Glandular elements minimal, lined by simple cuboid epithelium
- c. Abundant stroma consists of adipose tissue, fibroblasts

47. TESTIS

- a. Outer Tunica albuginea
- b. Sections of seminiferous tubules lined by spermatogenic cells and supportive Sertoli cells
- c. Interstitial cells of Leydig seen in between.

48. EPIDIDYMIS

- a. Tubules lined by pseudostratified ciliated columnar epithelium with stereocilia with clumps of spermatozoa within lumen
- b. Circularly arranged smooth muscle fibres and connective tissue around each tubule

49. VAS DEFERENS

- a. Mucosa, muscular layer, adventitia
- b. Mucosa thrown into folds lined by pseudostratified columnar epithelium
- c. Thick muscular coat with 3 layers – outer & inner longitudinal, middle circular

50. PROSTATE

- a. Prostatic acini lined by simple columnar epithelium
- b. Spherical prostatic concretions (corpora amylacea) inside follicles
- c. Fibro muscular stroma of connective tissue and smooth muscle fibres surrounding follicles

51. CEREBRUM

- a. Outer gray matter (cortex) and inner white matter
- b. Cortex shows 6 layers (superficial to deep) : Molecular layer, Outer granular layer, Pyramidal cell layer, Inner granular layer, Ganglionic layer, Polymorphous layer

52. CEREBELLUM

- a. Inner white matter and outer cortex
- b. Cortex made up of Outer molecular layer, Middle purkinje cell layer (Purkinje cells seen as flask shaped cells with ramifying dendrites), Inner granular layer

53. SPINAL CORD

- a. Inner "H" shaped gray matter with broad anterior horns (large multipolar neurons) and narrow posterior horns, central canal lined by ependymal cells
- b. Outer white matter in the form of posterior lateral and anterior funiculi made up of nerve fibers and neuroglia

54. CORNEA

- a. Stratified squamous nonkeratinised epithelium with anterior limiting membrane -Bowman's membrane
- b. Corneal stroma or substantia propria consisting of special collagen fibres and corneal corpuscles
- c. Descemet's membrane (posterior limiting membrane) with corneal endothelium lined by low cuboidal cells.

55. RETINA

- a. 10 layers from outer to inner
 1. Pigment epithelium
 2. Layer of Rods & Cones
 3. Outer limiting membrane
 4. Outer nuclear layer
 5. Outer plexiform layer
 6. Inner nuclear layer
 7. Inner plexiform layer
 8. Ganglion cell layer
 9. Optic nerve fibre layer
 10. Inner limiting membrane

56. THYROID GLAND

- a. Lobules made up of thyroid follicles lined by simple cuboidal epithelium and filled with homogenous colloid

- b. Para follicular cells between follicles as well as in follicular wall between basement membrane and follicular cells

57. PITUITARY GLAND

- a. Pars anterior showing chromophils (acidophils and basophils), chromophobes and sinusoids
- b. Pars posterior showing nerve fibres and pituicytes
- c. Pars intermedia shows colloid filled follicles

58. SUPRARENAL GLAND

- a. Outer cortex, inner medulla with numerous sinusoids
- b. Cortex has 3 zones – zona glomerulosa, zona fasciculata and zona reticularis.
- c. Medulla shows large chromaffin cells and sympathetic ganglion cells

59. TRACHEA

- a. Mucosa, submucosa, hyaline cartilage and adventitia
- b. Mucosa lined by pseudostratified ciliated columnar epithelium with goblet cells
- c. Hyaline cartilage as C shaped rings with trachealis(smooth) muscle at places where cartilage is absent.

60. LUNG

- a. Alveoli lined by simple squamous epithelium
- b. Bronchus lined by pseudostratified ciliated columnar epithelium surrounded by irregular cartilage plates and lamina propria with serous and mucous acini.
- c. Bronchiole lined by low cuboidal or pseudostratified ciliated columnar epithelium surrounded by prominent smooth muscle fibres, cartilage plates are absent